

TL Corp CRM Portal v20.0

FINAL Master Specification (Offline + Timeline + Performance Hardened)

Date: 2026-02-25

This version supersedes all prior versions. It incorporates architectural hardening focused on field reliability, enterprise clarity, and measurable performance standards.

1. Offline Field Mode (Required Implementation)

The system must support reliable jobsite usage in low or no connectivity environments.

- Offline capture allowed for: Action Items, Photos, Daily Reports, Time Entries.
- Offline entries stored locally with visible 'Pending Sync' indicator.
- Automatic background sync when connection restored.
- Conflict resolution rule: server version prevails unless user manually resolves conflict.
- User must see clear sync status icon (Green = Synced, Yellow = Pending, Red = Error).
- No data silently lost due to connectivity issues.

2. Project Activity Timeline (Enterprise Clarity Layer)

Each Project Workspace must contain a unified chronological activity feed.

- Timeline includes: Assignments, Action Item changes, Photo uploads, Daily Report submissions, Document views, Change Order events, Invoice events, Status changes.
- Events sorted newest to oldest with timestamps and user names.
- Filter by event type (e.g., Only CO events).
- Each timeline entry links to source object.
- Timeline must be exportable as part of Dispute Defense Packet.

3. Performance & Responsiveness Guarantees

Performance is a competitive differentiator. The following targets are required:

- Today Dashboard render under 2 seconds on standard 4G.
- Project Workspace initial load under 2 seconds.
- Action Item creation perceived response under 1 second.
- Photo compression before upload to reduce latency.
- Lazy load image galleries.
- Skeleton loaders for perceived speed improvement.

4. Stability & Data Protection Standards

- All uploads stored with retry logic.
- All background jobs (notifications, AI, exports) use queue system.
- Automatic nightly database backups.
- Exportable full project archive.
- Server-side validation on all create/update operations.

TL Corp CRM Portal v18.0

FINAL CONSOLIDATED IMPLEMENTATION SPECIFICATION

Date: 2026-02-25

This document replaces all prior versions. Use this specification as the sole source of truth. If any previous PDF conflicts with this version, this version prevails.

Terminology Lock (Authoritative)

The following terms are official and must be used consistently across UI, database, and documentation:

• Project Workspace (primary project detail screen; replaces any prior 'Job Bible' references) • Action Items (replaces Task / Work Item terminology) • Daily Report (replaces Daily Log) • Project Documents • Project Photos • Time Entries • Work Orders (reserved exclusively for Maintenance division service tickets) • Admin Console • Today Dashboard • Client Portal

All references to 'Task', 'Work Item', or 'Job Bible' in prior documents should be interpreted as 'Action Item' and 'Project Workspace' respectively.

TL Corp CRM Portal v16.0

SUPERNOVA Master Specification (Single PDF Hand-Off)

Date: 2026-02-23

Use this document instead of all previous PDFs. This file contains (1) market-benchmark guidance + simplification rules, then (2) the complete detailed specs from v12 + v13 + v15 appended in full.

Replacement note for the web developer: Please treat v16 as the source of truth. If any prior PDF conflicts, v16 wins.

1. Market Benchmark Findings (What the top systems do well)

These patterns are consistent across leading platforms (Procore for commercial execution, Buildertrend for residential client portal + updates, Jobber for service/maintenance operations, JobTread for budget→change order→job costing flow).

- Centralized hub: schedules, change orders, daily logs, punch/tasks connected as a single source of truth.
- Field speed: quick capture from the field (tasks/punch/photos) and offline-tolerant behavior.
- Client calm: client portals emphasize transparency without exposing internal noise.
- Service velocity: recurring visits + work orders + checklists for maintenance operations.
- Profit protection: change orders derived from budget items to avoid duplicate entry and reduce mistakes.
- Mobile parity: most critical workflows must work on phone in seconds.

Reference sources (for patterns only)

- Procore Project Management overview + Punch List: <https://www.procore.com/project-management> and <https://www.procore.com/project-management/punch-list>
- Buildertrend Client Portal + Project Management + Mobile Change Orders: <https://buildertrend.com/communication/construction-client-portal/> and <https://buildertrend.com/project-management/> and <https://buildertrend.com/help-article/change-orders-on-mobile/>
- Jobber Features (Scheduling, Client Hub, Forms/Checklists, Dispatching): <https://www.getjobber.com/features/>
- JobTread Change Orders from budget + Customer Portals/Job Costing: <https://www.jobtread.com/features/change-orders> and <https://www.jobtread.com/features/customer-portals> and <https://www.jobtread.com/features/job-costing>

2. The Simplification Doctrine (Same results, fewer taps)

To avoid Procore-level complexity while achieving Procore-level outcomes, the portal must hide complexity behind templates, defaults, and role-based views.

- **Keep Employee UI stable:** Today → Job Bible → Tasks/Photos/Time/Daily Log. No extra top-level modules for the field.
- **Make Admin the complexity sink:** RFIs/Submittals/Budget/Publishing/Reporting live in Admin and feed the field as simple tasks/checklists.
- **Client sees curated output only:** Updates, Published photos/docs, Selections (residential), CO approvals, Billing.
- **Templates everywhere:** Divisions/portfolios auto-load tasks, checklists, required photos, closeout gates.
- **One object model:** Projects + Work Orders share consistent patterns; maintenance is not a separate app—just a division/portfolio + templates.
- **Never require training calls:** Tooltips, guided tours, microcopy headers, confirmations, undo, and drafts.

3. Division + Portfolio Architecture (Replace 5 apps with 1 portal)

The system uses one data core and adds **Division** and **Portfolio** as context. Portfolios are primarily for Maintenance (Bonan Towers + future contracts).

Context	What it controls	What it must NOT change
Division (Residential/Commercial/Maintenance)	Enabled modules, default workflows, Capex budgets, etc.	Reporting, etc.
Portfolio (Bonan Towers, Future Maintenance Contracts, etc.)	Recurring schedules, asset/location lists	Division identity, base permissions model

4. Minimum Gap-Closers to Match the Top Dogs (without bloat)

- **Residential:** Selections/Allowances + Warranties/Callbacks (client approvals, status tracking).
- **Commercial:** RFI Lite + Submittal Lite + Drawings/Plans (latest) + Inspections/Observations (templated).
- **Maintenance:** Work Orders + Client requests + Recurring visits + Forms/Checklists library.
- **Profit:** Budget/Cost Codes Lite + Change Orders built from budget items.

5. How to Use This Document (Implementation map)

- Section A (front matter) gives the market-benchmark direction and simplification rules.
- Appendix 1 (v12) is the canonical functional+technical spec: data model, RBAC, events, API shapes, acceptance tests.
- Appendix 2 (v13) is the solo-dev playbook: screen-by-screen buttons, validations, toasts, error states, wire layouts.
- Appendix 3 (v15) is the benchmark upgrade addendum: divisions/portfolios + module activation + the missing ROI modules.

Developer instruction: build Phase 1 exactly to the Employee/Client/Admin core flows first, then add the benchmark modules per v15 without changing the employee shell.

TL Corp CRM Portal v12.0

FULL Functional + Technical Specification (SRD/PRD Hybrid)

Date: 2026-02-17

Purpose: Provide a no-ambiguity, build-ready specification for the TL Corp Digital Operating System. This document defines required behavior down to screen, button, validation, event, and audit level, and includes a recommended technical architecture (data model + API expectations) suitable for replacing paper workflows, supporting dispute defense, and scaling into commercial/enterprise use.

Prototype reference: crm-tl.vercel.app

0. Non■Negotiables (Definition of Success)

- Post-login behaves like an app (SPA): instant transitions, persistent navigation, no full page reloads.
- RBAC enforced server-side: clients see only their jobs; employees see only assigned jobs; admins control all.
- 3■Tap Rule for field actions from Employee Today: Navigate, Open Job, Clock In/Out, Add Photo, Add Task.
- Auto-save drafts everywhere + Resume Draft.
- Audit trail is immutable for critical events (schedule/scope/approvals/publishing/deletes).
- Electronic signatures with version lock + signed PDF snapshot.
- Document tracking events: Sent/View/Sign with notifications; events tied to document version.
- Replace paper-based workflows completely (daily logs, approvals, closeout packets).
- Usability: no user should need to call to learn flows; in-app help + tooltips + microcopy required.

1. Product Scope & Portal Breakdown

One system, one database, three portal shells: Employee, Client, Admin. Each role gets a tailored UI/permissions view of the same underlying data.

- Employee Portal (mobile-first): Today/Week, Job Bible (Tasks/Photos/Docs/Daily Log/Time/Messages), Fast field capture, acknowledgments.
- Client Portal (calm + curated): Overview, Updates, Photos (published only), Documents, Change Orders, Billing/Payments, Messages.
- Admin Portal (desktop-first): Dashboard, Jobs, Schedule/Dispatch, Publish Queue, COs, Invoices, Users/Roles, Templates, Reports.

2. Branding + Design System (Tokens are mandatory)

Implement tokens once and consume them across all components. Never hardcode per-component colors/spacing.

- Logo: fixed top-left in app shell; never stretched; maintain aspect ratio; clearspace $\geq$ height of the 'T' in TL mark.
- Primary TL Blue: #0B2A4A | Accent Gold: #C7A252 | Success #16A34A | Warning #F59E0B | Danger #DC2626
- Typography: Inter preferred; fallback system-ui.
- Spacing: 8pt grid (4/8/12/16/24/32/48/64).
- Tap targets: 44×44px minimum; button height 44–48px; input height 44px.
- Component set: AppShell, TopBar, Card, Button variants, Badge, Modal, Drawer, Toast, Skeleton, EmptyState, Stepper/Wizard, FileUpload, Table.

2.1 UI State Matrix (every component)

- Default, Hover (desktop), Pressed, Focus ring (2px #60A5FA), Disabled, Loading, Error, Success (where applicable).
- Every primary action must produce: visible state change + success toast + (if relevant) audit event.

3. Recommended Technical Architecture

This defines a reference architecture your dev can implement with Next.js/React + API routes/server + Postgres + object storage. The key is: server-side RBAC, immutable audit logging, and reliable event tracking.

- Frontend: SPA shell (Next.js app router or similar), role-based route guards; responsive layout.
- Backend: REST or GraphQL API; server-side authorization; background jobs for notifications (queue).
- Database: PostgreSQL (preferred) with UUID primary keys; row-level access enforced in service layer.

- Files: object storage (S3-compatible) for photos/docs; generate thumbnails; store metadata in DB.
- Search: simple DB search initially; optional full-text later.
- Observability: request logging + error tracking; audit export endpoints.
- Security: HTTPS only; secure sessions; optional 2FA for admin.

4. Data Model (Core Tables / Objects)

Minimum viable data model that supports all workflows. Field names below are canonical and should remain stable.

- User, Role, Permission, UserProjectAccess
- Project (Job), ProjectClientLink, Assignment (ScheduleItem), Milestone
- Task, TaskComment, TaskAttachment
- Photo, PhotoComment, PhotoPublishRequest
- Document, DocumentVersion, DocumentRecipient, DocumentEvent
- DailyLog, DailyLogSection, DailyLogAttachment
- TimeEntry, TimeEntryEditRequest
- ChangeOrder, ChangeOrderVersion, ChangeOrderSignature
- Invoice, Payment, Receipt
- MessageThread, Message, MessageAttachment
- AuditLog (immutable)

4.1 Project (Job) schema (required fields)

Field	Requirement
id	UUID
job_number	String; unique; format configurable (e.g., 26-014)
name	String; required
type	Enum: Residential / Commercial / Maintenance / Service / Warranty
status	Enum lifecycle: Draft→Scheduled→Assigned→Acknowledged→In Progress→Pending Review
address	Structured: street/city/state/zip + geo lat/long
access_notes	Text; optional
work_hours_allowed	Text/structured; optional
scope_summary	Text; required (admin)
today_target	Text; optional (employee-facing)

safety_notes	Text; optional
lead_user_id	FK User; required once Scheduled
created_by	FK User; required
created_at/updated_at	Timestamps

4.2 Assignment (ScheduleItem) schema

Field	Requirement
id	UUID
project_id	FK Project; required
start_at	Datetime; required
duration_minutes	Int; required
assigned_user_ids	Array FK Users; required
status	Enum: Draft / Assigned / Acknowledged / Completed
priority	Enum: Normal / Urgent / Emergency
critical_note	Short text shown on Today card; optional
changed_since_last_login_flag	Computed per user; drives banner
created_by/updated_by	FK User
created_at/updated_at	Timestamps

5. Roles & Permissions (RBAC)

RBAC must be enforced server-side. This table defines minimum permission expectations. 'R' = read, 'C' = create, 'U' = update, 'D' = delete, 'P' = publish.

Module	Client	Employee	Lead/Foreman	Admin
Projects/Jobs	R (own)	R (assigned)	R (assigned)	CRUD
Assignments/Schedule	R (own)	R (assigned)	R (assigned)	CRUD
Tasks	R (published opt.)	CRU (assigned)	CRU (assigned)+Assign	CRUD+Verify
Photos	R (published)	CR (internal)	CR (internal)	CRUD+Publish
Docs	R (published)	R (assigned)	R (assigned)	CRUD+Publish
Daily Logs	R (published summary opt.)	CRU (own draft)	CRU+Submit	CRUD+Export
Time	N/A	CRU (own)	CRU (crew view opt.)	CRUD+Approve
Change Orders	CRU (approve/decline)	R	R	CRUD+Send
Invoices/Payments	R+Pay	N/A	N/A	CRUD
Messages	R/W (admin thread)	R/W (job thread)	R/W	CRUD+Moderate
Audit/Reports	R (own exports opt.)	N/A	N/A	R+Export

6. Notification Engine (Required)

All important events generate in-app notifications. Email/SMS are optional channels. Notifications must deep-link to the exact object.

- Channels: In-app bell (required), Email (configurable), SMS (phase 2).
- Anti-spam: view events notify once per recipient (first view); subsequent views roll into activity feed or batched summary.
- Types: Assignment created/changed, Task assigned/blocked, Publish request, Doc viewed, Doc signed, CO approved/declined, Invoice paid, Daily log submitted, Safety acknowledgment.

6.1 Document tracking events (canonical)

- Created, Sent, Delivered(optional), Opened(optional email), Viewed(viewer), Downloaded(optional), Signed/Approved, Declined, Expired, Revoked.

6.2 AuditLog events (immutable)

- Schedule changes (old/new), job status changes, task status changes, publish actions, signature events, deletions/archives, reopen log with reason.

7. Employee Portal — Screen Specifications

All employee screens must be optimized for phones/iPads, low typing, big actions. Every screen includes a one-line 'What to do here' header and Help icon.

7.1 E1 Today (Default)

- Loads assignments for Today, sorted by start time.
- Job card fields: start time, job name/job#, address (truncate), crew, priority tags, critical note (1 line).
- Primary buttons per card: Navigate, Open Job, Clock In.
- Banner if updates since last login; tap opens Change Summary drawer listing changed fields.
- Long-press card opens quick actions: Add Photo, Add Task, Message Admin, View Docs.
- Empty state: 'No assigned work today' + buttons View My Week + Message Admin.

7.2 E2 My Week

- 7-day list, Today expanded by default; expand/collapse per day.
- Swipe assignment: Navigate, Message Admin.
- Filter toggle: All / Urgent / Emergency.

7.3 E3 Job Bible (Shell)

- Sticky Critical Info panel: Address (Navigate), Access notes, Scope summary, Today target, Safety/PPE, Contacts (tap-to-call).
- Tabs max 6: Tasks, Photos, Docs, Daily Log, Time, Messages.
- Back preserves prior scroll position.
- More menu: Request Materials, Report Issue, View Change Summary.

7.4 E4 Tasks (Tab)

- Sections: Open, In Progress, Blocked, Completed (collapsed).
- CTA: + New Task.
- Swipe left = Complete (confirm), swipe right = Block (confirm).
- Block requires blocked_reason dropdown + optional note/photo.
- Complete requires photo for categories QA/Safety/Punch (configurable rule).

7.5 E5 Task Detail

- Editable fields (if permitted): status, assignee, due date, notes, attachments.
- Comment thread with @mentions.
- Mini audit display: created by/at, last updated by/at.
- Actions: Duplicate (lead/admin), Cancel (lead/admin), Delete (time-window rule).

7.6 E6 Photos (Tab)

- + Photo opens camera; post-capture requires Tag (Before/During/After).
- Visibility defaults Internal; employee can Request Publish (creates admin queue item).
- Viewer supports comment; delete rules: uploader within 10 min or admin; Undo 10 min.

7.7 E7 Docs (Tab)

- Folders: Scope/Contract, Plans, Specs, Permits, Safety, Other.
- Latest versions default; toggle All versions read-only.
- PDF viewer: pinch zoom; optional search (phase 2).

7.8 E8 Daily Log (Tab)

- Status: Not started / Draft / Submitted.
- Sections: Work performed (template checkboxes + notes), Deliveries, Issues/Blocks, Safety, Client interaction.
- Unlimited photo attachments, tagged per section.
- Auto-save every 10 seconds; Submit locks; admin reopen requires reason (audit).
- If Issues entered → prompt: 'Create task from this?'

7.9 E9 Time

- Clock in/out by job; show running timer; Switch Job stops current and starts new (logs both).
- Break prompt at clock-out if missing.
- Edit requests allowed; admin approves.

7.10 E10 Messages

- Job thread + task comment threads; attachments internal unless published.

8. Client Portal — Screen Specifications

Client portal must be calm, curated, and non-technical. Clients see only published items.

8.1 C1 Overview

- Project status, milestone timeline, next steps.
- Action Items: Approve CO, Pay Invoice, Message Admin (conditional).
- Latest update preview.

8.2 C2 Updates

- Weekly/date cards; update detail includes summary + published photos + linked docs.

8.3 C3 Photos

- Published photos only; albums Before/During/After; viewer with optional download.

8.4 C4 Documents

- Contracts/estimates/CO PDFs; view-only by default; download configurable.

8.5 C5 Change Orders

- Pending first; CO detail shows scope/cost/time + disclaimer.
- Approve: e-sign typed+drawn; Decline requires reason; audit logged.

8.6 C6 Billing

- Invoice list + detail; Pay; receipt download; payment event logged.

8.7 C7 Messages

- Client↔Admin only; topic selector for new messages.

9. Admin Portal — Screen Specifications

Admin portal is the control center: job creation, scheduling, approvals, publishing, reporting.

9.1 A1 Dashboard

- Widgets: Today jobs, blocked tasks, publish requests, pending CO approvals, unpaid invoices, crew utilization.
- Widgets deep-link to filtered views.

9.2 A2 Jobs List

- Filters: status/type/lead/date range; row actions: Open, Schedule, Publish, Archive.
- Never hard-delete projects; Archive only.

9.3 A3 Create Project Wizard

- Step 1 Client+Address+Type; Step 2 Scope+Access+Hours+Safety; Step 3 Template; Step 4 Schedule+Crew; Step 5 Client visibility; Finish.
- Validation prevents missing required fields; finish triggers notifications.

9.4 A4 Schedule/Dispatch

- Create assignment; change assignment triggers employee 'Updated since last login' banner; optional acknowledgment required.

9.5 A5 Publish Queue

- Queues: photo publish requests, doc publish requests, update drafts; publish logs publisher+timestamp.

9.6 A6 Change Orders

- Create CO with scope/cost/time/disclaimer; send; track view/sign; approved version locked; optional invoice generation.

9.7 A7 Invoices & Payments

- Create/send invoices; track view/pay; receipt stored; audit logged.

9.8 A8 Users/Roles

- Invite users; assign roles; project access; disable retains audit history.

9.9 A9 Reports/Exports

- Dispute Defense Packet export (contract/COs/logs/photos/audit).
- Audit log export per project.
- Daily log export with embedded photos.

10. Legal Grade Workflows (Paper Replacement)

These workflows are required to make the portal legally defensible and commercial-ready.

10.1 Signature capture requirements

- Signature types: typed name + drawn signature; optional signature image upload.
- Metadata: signer name/email/role, timestamp, IP (optional), device info/user-agent (optional).
- After signature: generate locked PDF snapshot; store in immutable archive folder; version lock.

10.2 Daily log legal standard

- Unlimited photos; each stamped with user+timestamp; logs locked on submit; reopen requires reason + audit.
- Export PDF includes log text + embedded photos + timestamps.

10.3 Closeout & signoff

- Completion gate: required tasks/photos/checklists must be satisfied before marking Complete.
- Optional client on-site completion signature; warranty acknowledgment signature; punch signoff.

11. API Expectations (Reference Contract for Dev)

Developer may choose REST or GraphQL. Below is a recommended REST shape to eliminate ambiguity.

- **POST /auth/login | POST /auth/logout | POST /auth/register**
- **GET /me (current user + role + permissions)**
- **GET /projects?filters=... | POST /projects | GET /projects/:id | PATCH /projects/:id | POST /projects/:id/archive**
- **POST /assignments | GET /assignments/today | GET /assignments/week | PATCH /assignments/:id**
- **POST /tasks | GET /projects/:id/tasks | PATCH /tasks/:id | POST /tasks/:id/block | POST /tasks/:id/complete**
- **POST /photos (upload) | GET /projects/:id/photos | POST /photos/:id/publish-request | POST /photos/:id/publish**
- **POST /daily-logs/start | PATCH /daily-logs/:id | POST /daily-logs/:id/submit | POST /daily-logs/:id/reopen**
- **POST /time-entries/clock-in | POST /time-entries/clock-out | POST /time-entries/switch-job**

- **POST /change-orders | POST /change-orders/:id/send | POST /change-orders/:id/approve | POST /change-orders/:id/decline**
- **POST /invoices | POST /invoices/:id/send | POST /payments (webhook)**
- **GET /documents/:id/viewer (tokenized) | GET /documents/:id/events**
- **GET /audit?project_id=... | GET /exports/dispute-packet?project_id=...**

12. Acceptance Test Scenarios (QA Checklist)

These are executable test cases. Build is not complete until all pass on iPhone, iPad, and desktop.

- EMP-01: Employee opens app → Today loads within 2s → sees correct assignments.
- EMP-02: Employee clocks in from Today card → timer starts → toast appears → time entry created.
- EMP-03: Employee adds task → blocks it with reason → admin sees blocked task → audit entry exists.
- EMP-04: Employee submits daily log with photos → log locks → export PDF contains embedded photos.
- CLI-01: Client receives contract link → opens viewer → admin receives 'Viewed' alert.
- CLI-02: Client signs contract/CO → signed PDF snapshot stored → admin receives 'Signed' alert.
- ADM-01: Admin creates project via wizard → assigns schedule → employees notified → 'Updated since last login' banner works.
- ADM-02: Admin publishes photos → client sees them immediately, internal photos remain hidden.
- ADM-03: Dispute packet export generates bundle with contract, COs, logs, photos, audit timeline.
- SEC-01: Client cannot access another client's project via URL manipulation (RBAC enforced).

13. Delete/Archive Rules (Controlled Destruction)

To preserve legal defensibility, most objects are archived/cancelled rather than deleted.

- Projects: never hard-delete; Archive only.
- Tasks: creator may delete within 10 minutes; after that admin only; otherwise mark Cancelled (kept in audit).
- Photos: uploader may delete within 10 minutes; otherwise admin only; Undo 10 minutes.
- Docs: never delete versions; new upload creates new version; old versions remain read-only.
- Messages: never delete; allow admin redaction only with audit note (rare).

TL Corp CRM Portal v13.0

Solo-Developer Implementation Playbook (Buttons, Screens, Flows, States)

This playbook is written for a single developer. It removes ambiguity by defining: screen layouts, every button action, navigation behavior, validation rules, confirmation modals, toasts, error states, permissions, and minimum backend endpoints needed for each flow.

Reference prototype: crm-tl.vercel.app

1. Build Order (Do this in sequence)

- Step 1: Auth + RBAC routing shells (/app, /client, /admin).
- Step 2: Core components (Button, Card, Badge, Modal, Drawer, Toast, FormField, EmptyState, Skeleton).
- Step 3: Employee Today → Job Bible → Tasks → Photos → Time → Daily Log.
- Step 4: Admin Create Project Wizard → Schedule/Dispatch → Publish Queue.
- Step 5: Client Overview → Docs Viewer → CO Approve/Decline → Billing Pay.
- Step 6: Legal locks (signed PDF snapshots, version freeze) + AuditLog.
- Step 7: Document tracking events + notifications.
- Step 8: Exports (daily log PDF, dispute packet).

2. Global UI Rules (Applies to every screen)

- Every screen has: Title, 1-line 'What to do here' helper, Help icon.
- Every primary action produces: (1) UI state change, (2) success toast, (3) server event (and audit where relevant).
- All destructive actions require confirm modal with object name + 'Cancel' default focus.
- Auto-save drafts for forms every 10 seconds + 'Saved' indicator + Resume Draft on return.
- Back navigation preserves scroll position.
- Minimum tap target 44×44 px. No tiny icons without labels on mobile.

3. Button & Interaction Standards (Exact)

- Primary button label format: Verb + Object (e.g., 'Create Task', 'Submit Daily Log', 'Publish Photo').
- Confirm modals: 'Cancel' left, 'Confirm' right; destructive confirm is red and requires second confirmation only for irreversible actions.
- Toasts: top-right desktop, bottom center mobile; auto-dismiss 4s; include short action 'View' when applicable.
- Swipe actions (mobile lists): reveal 'Block' and 'Complete' on tasks; must show tooltip the first time user encounters swipe.
- Long-press (mobile): on job cards opens quick actions menu.

4. Screen Catalog (Routes)

This is the full minimum route set for v1 build.

- Public: /, /login, /register, /forgot-password

- Employee: /app/today, /app/week, /app/jobs, /app/jobs/:jobId, /app/messages, /app/time, /app/more
- Client: /client/overview, /client/updates, /client/photos, /client/documents, /client/change-orders, /client/billing, /client/messages
- Admin: /admin/dashboard, /admin/jobs, /admin/jobs/:jobId, /admin/schedule, /admin/publish-queue, /admin/change-orders, /admin/billing, /admin/users, /admin/templates, /admin/reports

EMPLOYEE PORTAL — Screen-by-Screen Button Specs

E1. Today (Default Home)

Purpose: Show the employee exactly what to do today and get them into the job in under 3 seconds.

Section	Details
Components	Header: 'Today' + date + Help (?) Banner (conditional): 'Updated since last login' (tap opens Change Sumr
Primary actions (buttons)	Tap job card → Open Job Bible (default tab: Tasks) Navigate → open device maps with address Clock
Validation / Rules	Clock In disabled if already clocked into another job unless user chooses Switch Job flow If job has critical
Success toasts	Clocked in on {Job#} at {time} Navigation started Opened job {Job#}
Error states	Schedule load error: 'Couldn't load today's schedule. Pull to refresh.' + Retry Clock-in error: 'Couldn't start t
Permissions	Employee: see only assigned jobs Lead: same, plus may see crew list (optional)
Navigation	Bottom nav: Today / Jobs / Messages / Time / More Back exits to previous screen; preserves scroll position

E2. My Week

Purpose: Let employee preview tomorrow/week without digging through calendars.

Section	Details
Components	Week list (Mon–Sun) with expand/collapse Assignment rows show time, job#, address, status pill Row
Primary actions (buttons)	Tap assignment → Open Job Bible Swipe row → reveal Navigate / Message Admin Tap day header –
Validation / Rules	Default expand Today; others collapsed If assignment is 'Assigned' and acknowledgment required, show 'A
Success toasts	Acknowledged assignment for {Job#} Navigation started
Error states	Week load error: 'Couldn't load week. Retry.'
Permissions	Employee: assigned only
Navigation	Back → Today

E3. Job Bible (Job Detail Shell)

Purpose: Single source of truth for the job: critical info + work modules.

Section	Details
Components	Sticky Critical Info panel (Address, Access, Scope, Today's Target, Safety, Contacts) Tabs: Tasks / Photos
Primary actions (buttons)	Tap Address → Navigate Tap contact phone icon → dialer Tab switch loads module instantly (no page transition)
Validation / Rules	If scope_summary empty (should not happen post-schedule), show admin warning banner (admin only) If scope_summary empty (should not happen post-schedule), show admin warning banner (admin only)
Success toasts	Copied address Started call (optional log prompt)
Error states	Job load error: 'Couldn't load job. Retry.'
Permissions	Employee: read-only job critical info; can create tasks/photos/logs/time Lead: may assign tasks to crew (photo)
Navigation	Back returns to the list that opened it (Today/Week/Jobs)

E4. Tasks Tab

Purpose: Create, track, block, complete all work items. Tasks are the field operating system.

Section	Details
Components	Sections: Open / In Progress / Blocked / Completed (collapsed) Floating '+ New Task' button Task row
Primary actions (buttons)	+ New Task → Create Task form Tap task row → Task Detail Swipe left → Complete (confirm) Swipe right → Block
Validation / Rules	Create Task requires Category + Title Block requires blocked_reason dropdown Complete requires 1 photo
Success toasts	Task created Task blocked: {reason} Task completed
Error states	Save error: 'Couldn't save task. Try again.'
Permissions	Employee: create/edit tasks they created; complete/block tasks they are assigned (configurable) Lead/Admin: create/edit tasks they created; complete/block tasks they are assigned (configurable)
Navigation	Back returns to Job Bible

E5. Daily Log Tab

Purpose: Fast, legally defensible daily documentation with photos and timestamps.

Section	Details
Components	Status: Not started / Draft / Submitted Buttons: Start Daily Log, Save Draft, Submit Sections: Work Performed, Notes, Photos
Primary actions (buttons)	Start Daily Log → creates draft for today Add Photo → camera/gallery → tag section Submit → confirm
Validation / Rules	Submit requires at least one Work Performed checkbox OR notes text Unlimited photos allowed If Issued checkbox checked, require photo
Success toasts	Daily log saved Daily log submitted
Error states	Submit error: 'Couldn't submit log. Try again.'
Permissions	Employee: create/submit own daily logs Admin: reopen with reason; export PDF
Navigation	Back returns to Job Bible

CLIENT PORTAL — Screen-by-Screen Button Specs

C1. Client Overview (Default)

Purpose: Show client progress, next steps, and required actions in a calm, curated view.

Section	Details
Components	Project status + milestone timeline Action Items banner (conditional) Latest update preview Navigation
Primary actions (buttons)	Tap Approve CO → opens CO detail Tap Pay Invoice → opens invoice detail Tap Message Admin →
Validation / Rules	Client sees only published photos/docs and their own invoices/COs If no updates, show empty state with e
Success toasts	Message sent Payment successful (after pay)
Error states	Load error: 'Couldn't load project. Please retry.'
Permissions	Client: read-only to project except approvals/payments/messages
Navigation	Bottom nav on mobile; sidebar on desktop

C2. Change Order Detail (Approve/Decline)

Purpose: Let client approve/decline with legally trackable e-signature and version lock.

Section	Details
Components	CO summary: scope, cost impact, time impact Attachments viewer Buttons: Approve, Decline E-s
Primary actions (buttons)	Approve → open e-sign modal → confirm → status Approved Decline → require reason note → confirm →
Validation / Rules	Approve requires typed name; drawn signature optional; timestamp captured After approval, generate lock
Success toasts	Change order approved Change order declined
Error states	Signature error: 'Couldn't capture signature. Try again.'
Permissions	Client can act only on their own COs
Navigation	Back to Change Orders list

ADMIN PORTAL — Screen-by-Screen Button Specs

A1. Create Project Wizard

Purpose: Ensure every job is created with all required information, templates, and assignments (no missing fields).

Section	Details
Components	Stepper: 1 Client/Address/Type, 2 Scope/Access/Hours/Safety, 3 Template, 4 Schedule/Crew, 5 Client visibility
Primary actions (buttons)	Finish → creates project + assignments + notifications Send Client Invite toggle at Finish
Validation / Rules	Finish disabled until required fields satisfied Job number auto-generated or entered; must be unique
Success toasts	Project created Client invite sent (if toggle on)
Error states	Create error: 'Couldn't create project. Check fields and retry.'
Permissions	Admin only
Navigation	After Finish → route to Admin Project Detail

A2. Publish Queue (Photos/Docs/Updates)

Purpose: Control what becomes client-visible. This protects TL Corp and keeps client portal calm.

Section	Details
Components	Tabs: Photo Requests, Doc Requests, Update Drafts Per-item actions: Publish, Keep Internal, Add Caption
Primary actions (buttons)	Publish photo → choose album (Before/During/After) + caption → Publish Keep internal → dismiss request
Validation / Rules	Publishing logs publisher + timestamp Client-visible flag flips only via Publish action
Success toasts	Published to client Kept internal
Error states	Publish error: 'Couldn't publish. Retry.'
Permissions	Admin only
Navigation	Deep-link from notifications to exact request

5. Wireframe Reference (ASCII layout snapshots)

These are layout references (not final art). Use as positional guidance for primary actions.

```
EMPLOYEE TODAY (MOBILE)
+-----+
| TL [Logo]           Today, Tue           |
| [Updated since last login] (tap for details) |
+-----+
| 7:30 AM  Job 26-014  1234 Clayton Rd      |
| Crew: Mike + Aaron  Tags: Urgent | Safety  |
| Note: Cabinet delivery—confirm access code |
```



```

| [Navigate] [Open Job] [Clock In] |
+-----+
| 10:30 AM Job 26-015 55 Bonan Towers |
| Crew: ... |
| [Navigate] [Open Job] [Clock In] |
+-----+
| Today | Jobs | Messages | Time | More |
+-----+

```

```

JOB BIBLE (MOBILE)
+-----+
| < Back   Job 26-014   (More ■) |
+-----+
| Address: 1234 Clayton Rd   [Navigate] |
| Access: Lockbox 1234 | Park rear |
| Scope: Bath remodel-tile + vanity + paint |
| Today: Set vanity + trim plumbing |
| Safety: PPE required (tap) |
+-----+
| Tabs: Tasks | Photos | Docs | Daily Log | Time |
+-----+

```

6. What to send the developer

- Send this v13 playbook + the v12 functional/technical spec (for data model + audit + API).
- Tell him: v13 defines buttons/flows; v12 defines architecture/data model/audit/events.
- If he wants visual references: use this doc's ASCII wireframes + review Procore/Fieldwire/Buildertrend patterns for inspiration.

TL Corp CRM Portal v15.0

Market Benchmark Upgrade Addendum (Procore / Buildertrend / Jobber / JobTread)

Date: 2026-02-23

Purpose: Upgrade the TL Corp portal spec to match and surpass leading platforms by adding missing high-ROI modules without bloating field UX. This addendum plugs directly into v12 (architecture/data model) and v13 (button-level playbook).

1. Division + Portfolio Layer (Replaces 5 apps without 5 systems)

Implement a global Division/Portfolio selector that changes templates, defaults, and enabled modules—but does NOT change the core data model.

- Global selector (Admin + Management views): Division = Residential / Commercial / Maintenance.
- Maintenance Portfolio selector: Bonan Towers + (future contract names) as dropdown.
- All records store: division_id + portfolio_id (nullable for Residential/Commercial).
- Employees do NOT choose division: they only see assigned work; division only changes templates/forms shown.

2. Benchmark Feature Gap Closers (High ROI, Low UX Cost)

- Residential: Selections/Allowances + Warranties module.
- Commercial: RFI Lite + Submittal Lite + Drawings/Plans + Inspections/Observations Lite.
- Maintenance: Work Orders / Service Requests + Recurring Visits + Forms/Checklists library.

3. Module Activation Matrix (What shows up where)

Keep employee UI minimal; put complexity in admin/management screens. The matrix defines module visibility per division.

Module	Residential	Commercial	Maintenance	Employee UI	Client UI
Selections/Allowances	Yes	No	No	Read-only prompts	Approve/choose
Warranties/Callbacks	Yes	Optional	Yes	Assigned tickets	Request/track
RFI Lite	No	Yes	Optional	Create question	View answer (optional)
Submittal Lite	No	Yes	No	View approved	View approved (optional)
Drawings/Plans	Optional	Yes	Optional	View latest	View published
Inspections/Observations	Optional	Yes	Yes	Run checklist	View results (optional)
Work Orders (Service)	Optional	Optional	Yes	Core module	Request work
Recurring Visits	Optional	Optional	Yes	Shows as assignment	See schedule (optional)
Budget/Cost Codes	Yes (lite)	Yes	Optional	Hidden	Hidden
CO from Budget Items	Yes	Yes	Optional	Hidden	Approve/sign

4. Residential Additions (Buildertrend-style, TL-simple)

Add two modules that reduce rework and client confusion while keeping UI clean.

- Selections/Allowances: category, item, link/photo/spec, status (pending/approved/ordered/received/installed), owner (client/admin), due date, notes.
- Client flow: Client Portal → Selections → approve/choose → timestamped approval (audit logged).
- Admin flow: create selection list from template → publish to client → track statuses → optional auto-create tasks on status changes.
- Warranties: warranty ticket, assign tech, schedule visit, capture photos, client signoff optional, close and archive.

5. Commercial Additions (Procore-style modules, lite versions)

Add commercial credibility modules with narrow scope: fast, searchable, tied to the job.

- RFI Lite: question + location + attachments + assignee + due date → response → status (open/answered/closed). Export RFI log (PDF/CSV).
- Submittal Lite: submit spec/product data + attachments → approve/reject → store approved version; link to tasks/COs.
- Drawings/Plans: foldered, versioned, 'Latest' default; markups phase 2; prevent building from old set.
- Inspections/Observations Lite: checklist templates + pass/fail + photos + corrective task creation; export report.

6. Maintenance Additions (Jobber-style speed)

Maintenance/Bonan behaves like service work: fast tickets, recurring visits, repeatable checklists.

- Work Order entity: request source, priority, category, location, assigned tech, scheduled visit(s), status lifecycle.
- Client Hub (maintenance clients): request work, approve quote (optional), view scheduled visits, pay invoice (optional).
- Recurring Visits: schedule generator (monthly/quarterly/annual) that creates assignments + templated checklists.
- Forms/Checklists library: portfolio-specific templates (e.g., mechanical room, sprinkler pump, elevator phone checks).

7. Job Costing Lite + CO-from-Budget (JobTread concept, TL-simple)

Add lightweight budgeting to power accurate change orders and profit protection without full accounting.

- Cost Codes starter set: Labor, Materials, Subcontract, Equipment, Permits/Fees.
- Budget Items: code, description, qty/unit, unit cost, planned total; optional vendor.
- Actuals: time entries map to labor code; receipts map to materials; vendor bills optional phase 2.
- Change Orders created by selecting budget items (no duplicate entry); CO shows added/removed items.
- Management dashboard: Planned vs Actual by project and by division.

8. Document Tracking Enhancements (Sales + Legal)

- Per-recipient tracking: viewed/signed status shown for each recipient/signer.
- Reminder engine: if viewed but not signed after X hours → alert admin + suggested follow-up message.
- Timeline view: Created→Sent→Viewed→Signed with timestamps; linked to version hash and signed PDF snapshot.

9. Implementation Notes (Solo dev friendly)

- Do NOT add top-level nav items for every module on employee UI. Keep employee nav stable; modules live inside Job Bible tabs or 'More'.
- Use division/portfolio templates to load: tasks, checklists, selections, warranties, inspection forms.
- Every module integrates with: AuditLog, Notifications, Search, Exports (where relevant).

TL Corp CRM Portal v17.0

FINAL BOSS – Developer Decision Authority + Clarification Addendum

Date: 2026-02-23

Purpose: Eliminate remaining interpretation gaps for a solo developer. This document defines default technical decisions, UI preferences, naming conventions, and authority rules when ambiguity arises.

1. Terminology Clarification – 'Job Bible'

'Job Bible' is an internal product name created in this specification. It refers to the primary Job Detail screen containing critical info and tabs (Tasks, Photos, Docs, Daily Log, Time, Messages).

- Public-facing label should be 'Job Details' (not 'Job Bible').
- Internal documentation may reference 'Job Bible' as architectural term only.
- Route: /app/jobs/:jobId

2. Developer Default Decision Rules

If the spec does not explicitly define behavior, follow these defaults:

- Use drawer over modal for multi-step content unless destructive action.
- Use modal for confirmations and irreversible actions only.
- Prefer tabs over nested sub-routes inside Job Details.
- Prefer optimistic UI updates with rollback on server failure.
- Never hard-delete core records (archive instead).
- All IDs must be UUID.
- All timestamps stored in UTC.
- All permissions enforced server-side, not client-side only.

3. UI Behavior Authority

- Buttons: Verb-first labels (e.g., 'Create Task', 'Submit Log').
- Confirm modals must include object name (e.g., 'Delete Task: Install Vanity?').
- Every primary action must produce success toast.
- All forms auto-save drafts every 10 seconds.
- Focus states required for accessibility.
- Mobile-first layouts take priority over desktop aesthetics.

4. Notification Engine Defaults

- In-app notification is primary channel.
- Email notifications enabled for signatures, invoices, and assignment changes.
- Batch multiple view events into single summary notification.

- Never send more than one signature reminder within 24 hours.

5. Legal Lock Rules

- Once signed, document version is immutable.
- Edits require version increment and new signature.
- Signed PDFs stored in separate immutable archive folder.
- Audit log cannot be edited or deleted via UI.

6. Division & Portfolio Enforcement

- Division controls template loading only.
- Portfolio controls recurring maintenance defaults.
- Division must NOT change employee navigation structure.
- New maintenance contracts must be created as Portfolio entries.

7. Final Acceptance Trigger

System is considered production-ready when:

- All v12 functional requirements pass.
- All v13 screen/button flows behave as specified.
- All v15 modules activate per matrix rules.
- No employee requires external instruction to complete a job cycle.
- Signed document export contains metadata and version hash.
- Admin can generate full dispute defense packet in under 30 seconds.

TL Corp CRM Portal v19.0

AI Assist Addendum (Admin + Employee + Client)

Date: 2026-02-25

1. AI Design Principles (Non-Negotiable)

- AI must assist, not replace workflows.
- AI outputs are always editable by humans before publishing/sending.
- AI suggestions are optional (Accept / Edit / Dismiss).
- All AI actions create a log entry (AIActionLog) for traceability.
- AI must never publish to clients automatically without admin confirmation.
- AI features must not add taps to core field workflows; they must reduce taps.

2. AI Modules (All Included)

- AI Weekly Update Draft Generator (Admin).
- AI Daily Report Assistant (Employee: voice-to-structured + suggestion).
- AI Scope Completeness Checker (Admin, before sending estimate/contract).
- AI Action Item Suggestor (Employee/Admin; convert notes/issues into Action Items).
- AI Dispute Summary Generator (Admin/Management).
- AI Command Assistant (search/chat inside portal).
- AI Photo Auto-Tagging + Caption Suggestions (Employee/Admin).

3. AI UI Insertion Points (Where it appears)

AI must appear as a consistent 'sparkle' button with label 'AI Assist'. Avoid hidden AI features.

- Admin → Updates: 'AI Draft Update' button.
- Employee → Daily Report: microphone button + 'AI structure' button.
- Admin → Project Setup / Scope: 'AI Check Scope' button.
- Employee/Admin → Notes fields: 'AI → Create Action Items' button.
- Admin → Reports: 'AI Dispute Summary' button.
- Global top bar: 'AI Assistant' icon opens chat drawer.
- Photos: 'AI Suggest Tags' on upload screen.

4. AI Weekly Update Draft Generator (Admin)

- Input sources: last 7 days Daily Reports, published photos, completed Action Items, CO approvals, milestone changes.
- Output: draft update with sections: Summary, Work Completed, Upcoming Work, Decisions Needed, Risks/Delays.
- Admin can Edit, Publish, or Discard.
- Publishing creates Update record and client notification; AI output is never auto-published.

5. AI Daily Report Assistant (Employee)

- Voice-to-structured: employee speaks; AI fills Work Completed, Issues/Blocks, Safety notes fields.
- Auto-suggest Action Items from Issues/Blocks with Accept/Reject buttons.
- AI must preserve raw transcript as optional reference (internal only).
- Employee can edit before Submit; Submit locks report per legal rules.

6. AI Scope Completeness Checker (Admin)

- Triggered before sending estimate/contract/CO.
- Input: scope summary text + selected templates + division type.
- Output: checklist of commonly missed items (ex: protections, disposal, permits, punch/cleanup, warranty, exclusions).
- Admin chooses: Add items, Ignore (with reason), or Save as note.
- All ignores with reason logged for internal QA.

7. AI Action Item Suggestor (Admin/Employee)

- Triggered from any notes field: Issues/Blocks, Daily Report notes, messages.
- Output: proposed Action Items with Category, Priority, Location, Assignee suggestions.
- User can Accept all, Accept selected, Edit, or Dismiss.
- Accepted items create Action Items with 'created_by = AI Assist' + human approver stored.

8. AI Photo Auto-Tagging + Captions

- On upload, AI suggests: Before/During/After, Location tags, short caption.
- User can accept/edit.
- If AI confidence low, show 'Suggestions' as optional only.

9. AI Dispute Summary Generator (Admin/Management)

- Input: signed contract versions, COs, Daily Reports, Photos, Document tracking events, Audit log timeline.
- Output: internal narrative summary + key dates + supporting links.
- Must include citations to record IDs/timestamps (internal), not external legal claims.
- Export can be attached to Dispute Defense Packet (internal).

10. AI Command Assistant (In-App Chat)

- Chat drawer supports queries like: 'Show overdue Action Items', 'Summarize Project 26-014', 'Draft follow-up message'.
- Assistant can generate links to filtered views (not make silent changes).
- Any action that changes data requires a confirm screen.
- Assistant logs query + output reference in AIActionLog.

11. AI Logging + Governance (Required Tables)

- AIActionLog: id, user_id, project_id, feature_type, input_refs, output_text, accepted_boolean, created_at.
- AIFeatureFlags: enable/disable per role and per division.
- Privacy: do not include sensitive client information in AI prompts beyond what is necessary.
- Data retention: AI logs retained with project retention policy; exportable for internal audit if needed.

12. Acceptance Criteria (AI)

- AI features never increase tap count in core workflows; they must reduce manual typing.
- AI never auto-publishes to clients without admin confirm.
- Every AI output is editable before submit/publish.
- AI suggestions can be dismissed without penalty and do not block progress.
- AIActionLog records are created for each AI interaction.
- AI Dispute Summary references actual system events accurately (IDs/timestamps).